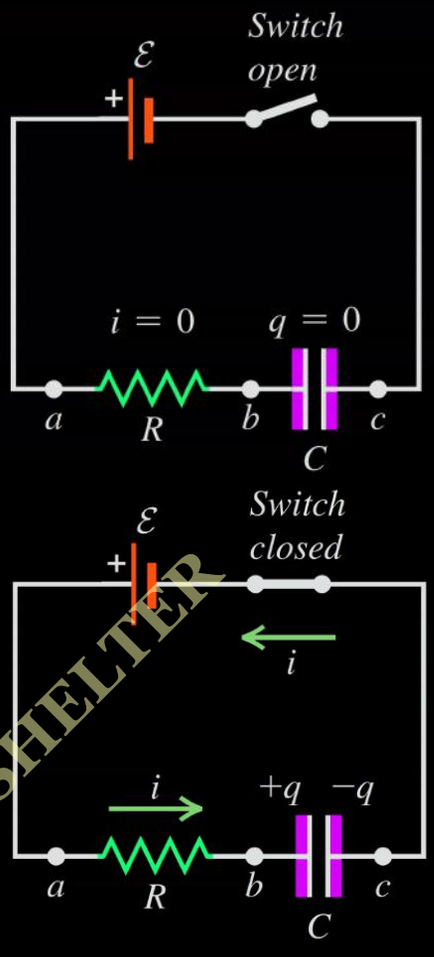
Charging An Uncharged Capacitor:

Consider a simple series RC circuit as shown in adjacent figure. Initially the capacitor is uncharged and the switch is open and therefore no current flows. When the switch is closed, the initial potential difference across capacitor is zero, thus P.D. across resistor is $\mathcal{E}$ and an initial current of $\mathcal{E}/R$ begins to flow, as the charge on capacitor increases, the potential drop across it increases and hence that across resistor decreases, thus current keeps decreasing as charge on capacitor keeps increasing.



Now applying KVL to the circuit:

$$iR + q/C - \mathcal{E} = 0, \text{ but } i = dq/dt, \text{ thus:}$$

$$R\,dq/dt = (\mathcal{E} - q/C) = (C\mathcal{E} - q)/C \Rightarrow dq/(C\mathcal{E} - q) = -dt/RC$$

Now integrating:

$$\int_0^q \frac{dq}{C\mathcal{E} - q} = -\frac{1}{RC}\int_0^t dt,$$

Now integrating and rearranging:

$$q = C\mathcal{E}(1 - e^{-t/RC})$$

This can also be written as:

$$\boxed{q = q_\infty(1 - e^{-t/\tau})}$$

Here $q_\infty = C\mathcal{E}$ i.e charge at $t = \infty$,

$\tau = RC$ also known as time constant which also the time for current to fall to $1/e$ times of its initial value.

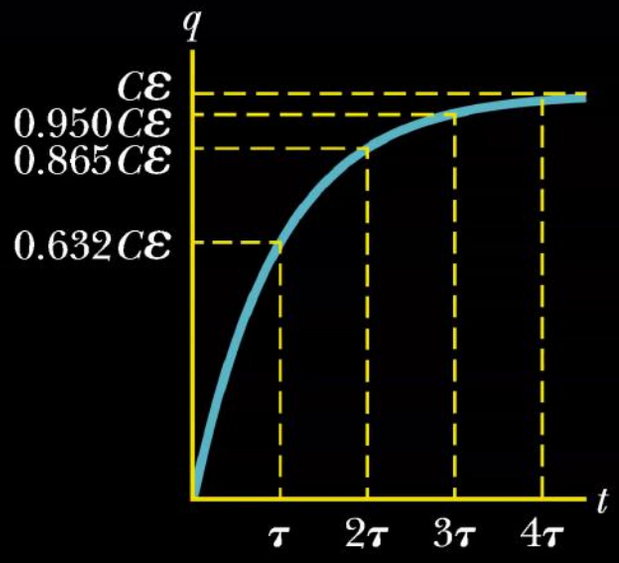
Now we can also find current as a function of time by using $i = dq/dt$. Using this we get:

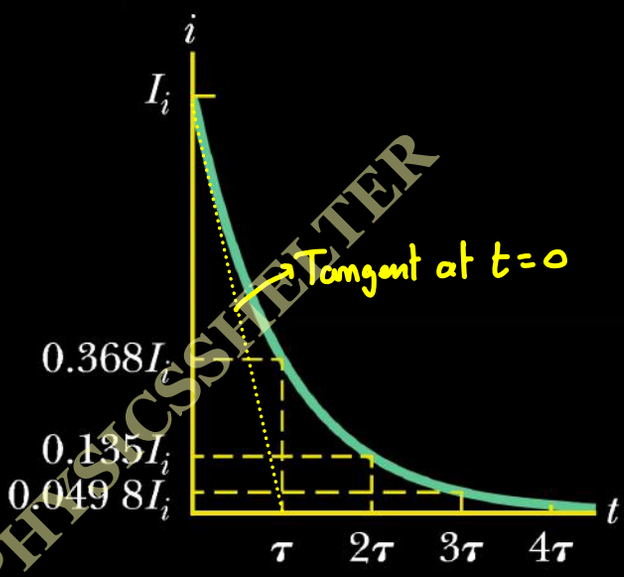
$$i = \frac{\mathcal{E}}{R}e^{-t/RC},$$ this can also be written as $\boxed{i = i_0 e^{-t/\tau}}$

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Here $i_0 = \mathcal{E}/R$ i.e current at the initial moment.

→ Initial current in the circuit is as if the capacitor were replaced by a short circuit while at $t = \infty$, the current is zero, i.e as if capacitor is replaced by an open circuit. Thus we say that initially (an uncharged) capacitor behaves like a short circuit and in steady state capacitor behaves like an open circuit.





The graphs of charge vs time and current vs time are plotted above, one can see that even though theoretically it will take infinite time to charge a capacitor, however practically the complete capacitor is fully charged within a few time constants. (86.5% in two time constants and 95% in three time constants). In one time constant, the capacitor is about 63.2% charged. Similarly, current decays to zero theoretically in an infinite time, but practically it becomes very small within a few time constants. (within 3 time constants, decays to 5% of its initial value, within 2 time constants decays to 13.5% and in one time constant decays to 1/e times i.e 36.8% of its initial value.)

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VOLTAGE VARIATIONS DURING CHARGING A CAPACITOR:

Voltage across capacitor $V_c = q/C = \mathcal{E}C(1 - e^{-t/RC})/C$

$\Rightarrow$

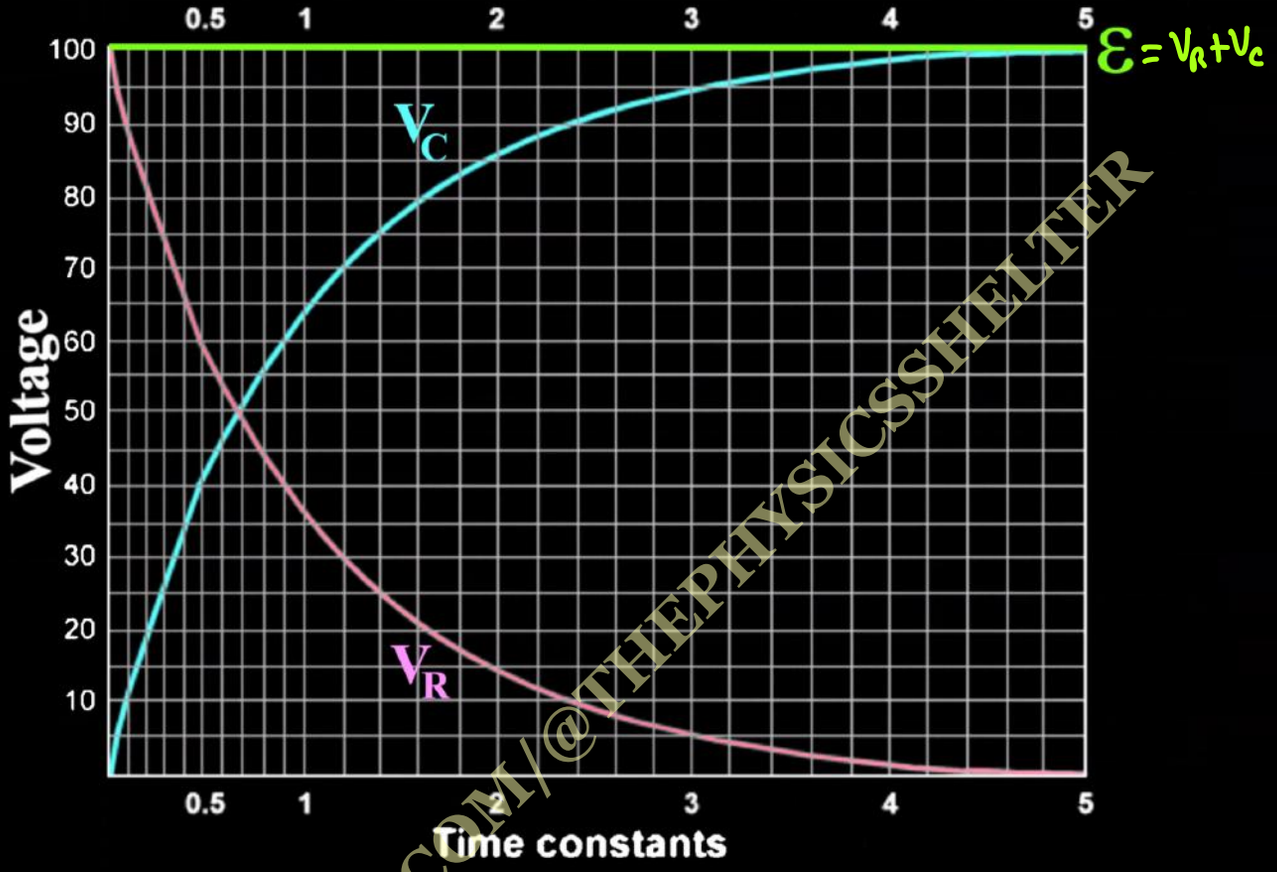
$$\boxed{V_c = \mathcal{E}(1 - e^{-t/\tau})}$$

Voltage across resistor: $V_R = iR = \frac{\mathcal{E}}{R} e^{-t/RC} \times R = \mathcal{E} e^{-t/\tau}$

Thus

$$\boxed{V_R = \mathcal{E} e^{-t/\tau}}$$

Note that $V_C + V_R = \mathcal{E}(1 - e^{-t/\tau}) + \mathcal{E} e^{-t/\tau} = \mathcal{E}$ = constant. (They must add up to EMF of the battery). The following graph shows variation of these voltages with time:



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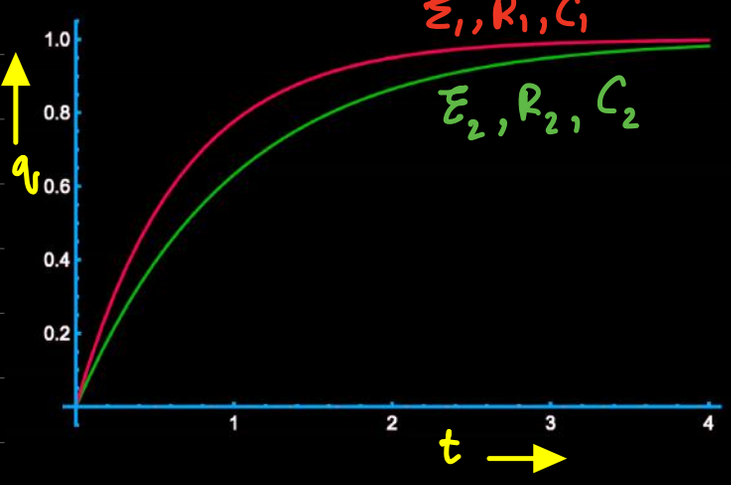
EXAMPLE: Adjacent figure shows charging graphs of two different RC circuits. Saturation charge for both is same. It can be concluded that:

(a) $\mathcal{E}_1 C_1 = \mathcal{E}_2 C_2$

(b) $R_1 C_1 > R_2 C_2$

(c) $C_1 > C_2$

(d) $R_1 > R_2$

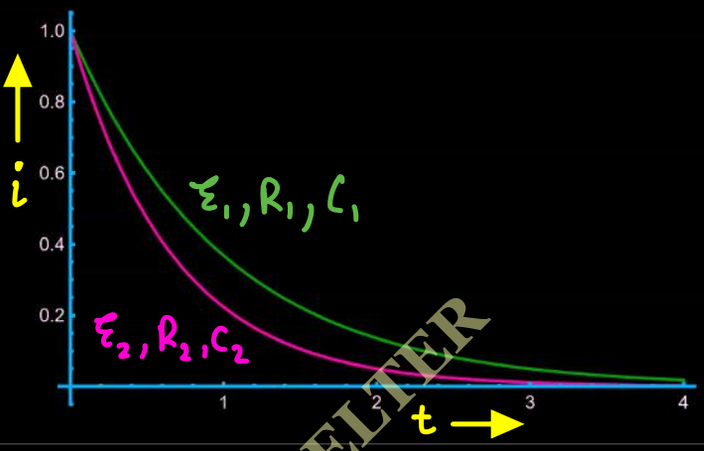


Solution: $q = q_0(1 - e^{-t/RC})$, Now q_0 is same for both thus,

$\mathcal{E}_1 C_1 = \mathcal{E}_2 C_2 \Rightarrow$ a is correct ($\because q_0 = \mathcal{E}C$)

also larger the time constant, longer it takes to saturate thus, $R_1C_1 < R_2C_2$ (b is not correct), and we can't comment of (c) and (d) options unless more information is available on $\mathcal{E}_1$ and $\mathcal{E}_2$. (For instance if $\mathcal{E}_1 < \mathcal{E}_2 \Rightarrow C_1 > C_2 \Rightarrow R_1 < R_2$)

EXAMPLE: Current vs time graph for two charging RC circuits are as shown in the figure. Initially capacitors had zero charge in both circuits then:



(a) $\mathcal{E}_1/R_1 = \mathcal{E}_2/R_2$

(b) $R_1C_1 > R_2C_2$

(c) $R_1 > R_2$

(d) $C_1 > C_2$

Solution: Since initial current is same for both circuits thus, $\mathcal{E}_1/R_1 = \mathcal{E}_2/R_2$. Also $i = (\mathcal{E}/R)\,e^{-t/RC}$, thus $R_1C_1 > R_2C_2$ (Smaller time constant $\Rightarrow$ Quicker decay of current). Thus options (a) and (b) are correct but to comment on (c) and (d) we need more information. (For instance if $\mathcal{E}_1 < \mathcal{E}_2$ then $R_1 < R_2$ and then C_1 will have to be greater than C_2).

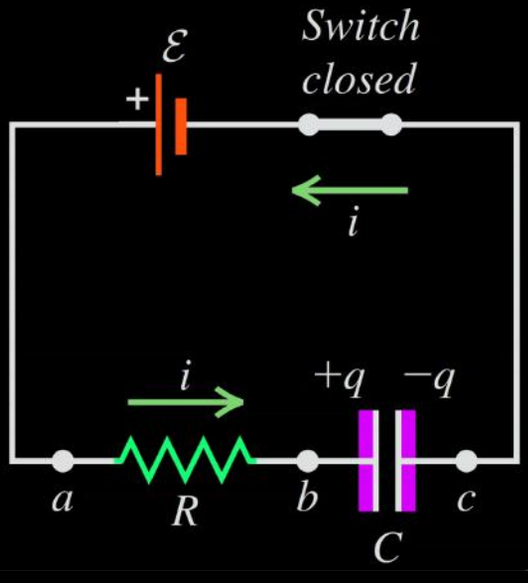
ENERGY STORAGE AND DISSIPATION WHILE CHARGING A CAPACITOR:

Consider the circuit shown in the adjacent figure. Applying KVL: $iR + q/C - \mathcal{E} = 0$

Rearranging: $\mathcal{E} = iR + q/C$

Now multiplying throughout by i:

$\mathcal{E}i = i^2R + iq/C$, This is simply equation of energy conservation, here LHS represents rate of work done by battery, first term on RHS represents rate of heat dissipation and second term on RHS represents Rate of Energy storage in capacitor.



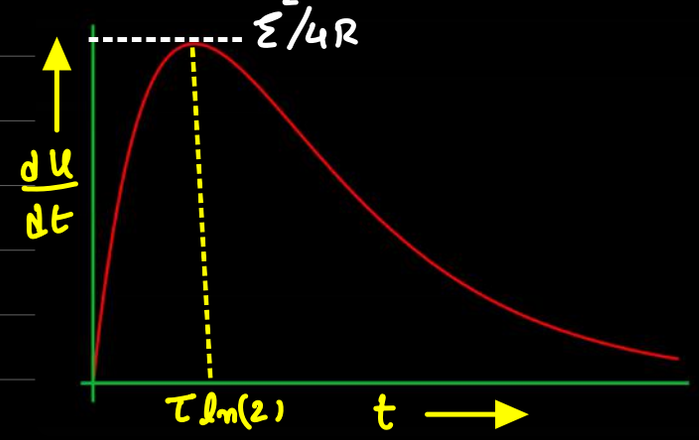
Now rate of storage: $\frac{dU}{dt} = \frac{iq}{C}$

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Thus $\frac{dU}{dt} = 0$ at $t=0$ and $t=\infty$

($\because$ @ $t=0$, $q=0$ and @ $t=\infty$, $i=0$)

Now we might want to know when is rate of storage of energy maximum, for this we need to maximize the product $i \times q$ ($\because v = iq/C$) i.e we need to maximize $i_0 e^{-t/\tau} \times q_\infty (1-e^{-t/\tau})$, Now if we put $e^{-t/\tau} = x$, the problem reduces to maximizing $x(1-x)$, which we readily know from A.M. > G.M that it occurs for $x = 1/2$. Thus for maximum rate of energy storage, $e^{-t/\tau} = 1/2 \Rightarrow$ $\boxed{t = \tau \ln 2}$



and corresponding value of $\frac{dU}{dt} = \frac{i_0 q_\infty}{4C} = \frac{\mathcal{E}^2}{4R}$

Now Heat Dissipated till any time can be easily found using first law of thermodynamics i.e $\Delta H = \Delta W_{bat} - \Delta U$ or alternatively integrating i^2R. Using first approach:

$$H(t) = \mathcal{E} q(t) - q^2/2C = \mathcal{E}\left[\mathcal{E}C(1-e^{-t/RC}) - \frac{[\mathcal{E}C(1-e^{-t/RC})]^2}{2C}\right]$$

$$= C\mathcal{E}^2(1-e^{-t/\tau})\left[1 - \frac{1}{2}(1-e^{-t/\tau})\right]$$

$$= \frac{C\mathcal{E}^2}{2}\left[1 - e^{-2t/\tau}\right] \Rightarrow \boxed{\Delta H = \frac{1}{2}C\mathcal{E}^2(1-e^{-2t/\tau})}$$

Now using second method, $\Delta H \int i^2 R\,dt = \int_0^t \left(\frac{\mathcal{E}}{R} e^{-t/RC}\right)^2 R\,dt$

$$= \frac{\mathcal{E}^2}{R}\left[\left(e^{-2t/RC}\right)\right]_0^t \times \left(\frac{RC}{-2}\right)$$

$\Rightarrow$ $\boxed{\Delta H = \frac{1}{2}C\mathcal{E}^2(1-e^{-2t/\tau})}$, At $t=\infty$, $\Delta H = \frac{1}{2}C\mathcal{E}^2 = \Delta U$

$\rightarrow$ Thus during complete charging of a capacitor from zero charge,

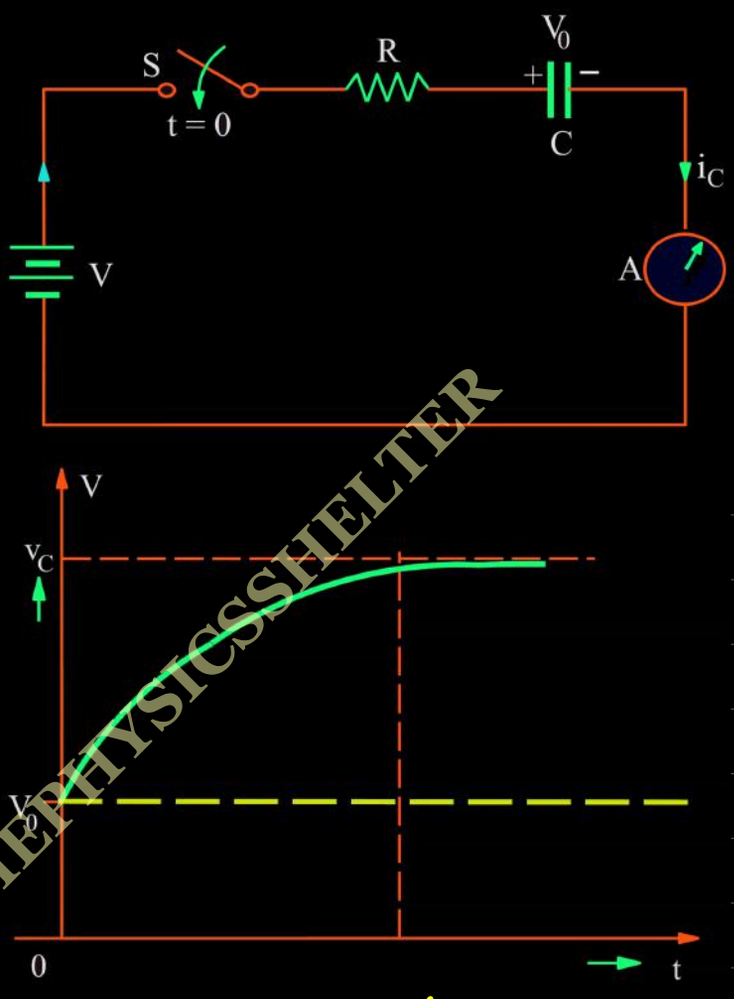
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exactly 50% of energy is dissipated and 50% is stored.

Further Charging of Precharged Capacitor:

Let a capacitor be precharged to a potential V_0 with polarity as shown, now let x be the additional charge flown from battery at a general time after closing the switch, then KVL for the loop gives

$$iR + \frac{x}{C} + V_0 - V = 0$$



This differential equation is as if a battery of $V - V_0$ were connected for purpose of calculating extra charge, thus we can readily write following solutions for the circuit:

$$q = q_0 + (q_\infty - q_0)\left[1 - e^{-t/\tau}\right]$$

→ Considering Yellow line as base, $V_{eff} = V - V_0$

Here $q_\infty = CV$, The second term on RHS represents additional charge flown. Similarly for current we can write:

$$i = i_0 e^{-t/\tau} \quad \text{where} \quad i_0 = (V - V_0)/R$$

For voltage on capacitor we can write:

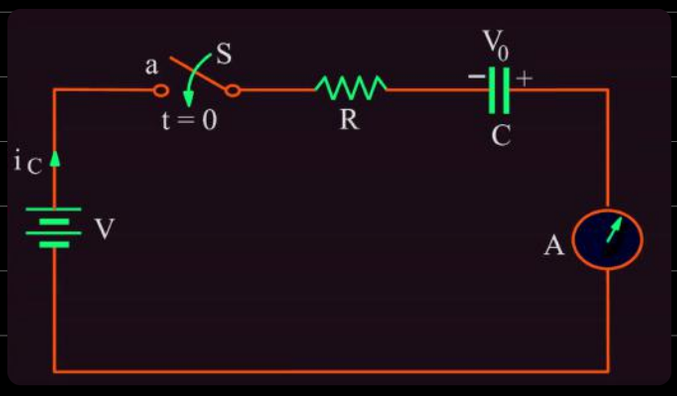
$$v_c = V_0 + (V - V_0)\left[1 - e^{-t/\tau}\right]$$

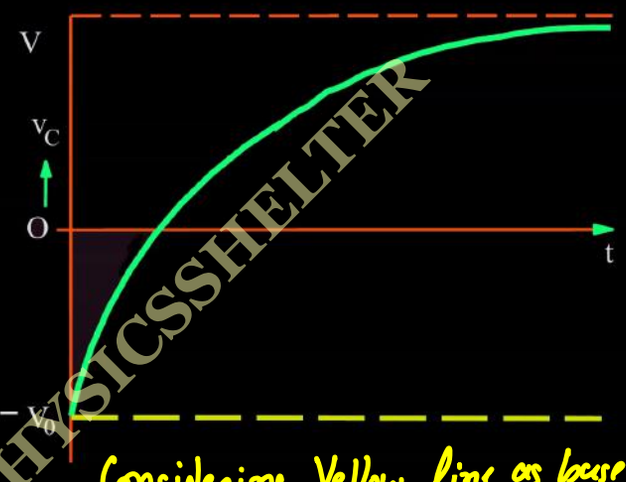
Here second term on RHS represents additional potential difference. Similarly for potential difference across resistor we can write:

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$$U_R = (V - V_0)\, e^{-t/\tau}$$

Note that in above analysis, V_0 is a signed value, for instance if capacitor had an initial polarity as shown in adjacent figure, then V_0 will have to be substituted by $-V_0$.
An easy way of dealing with such problems will be to recognise the value of effective EMF for purpose of calculating additional charge flow and then superpose original charge to find the net charge and net potential difference.

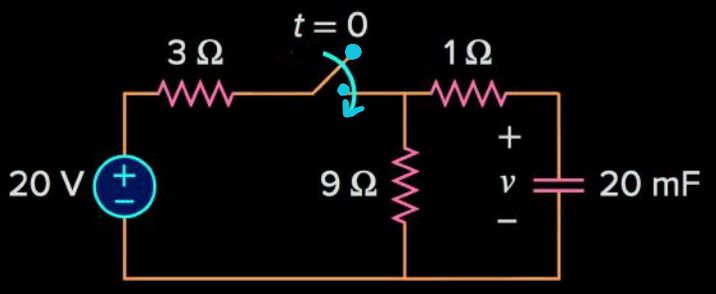




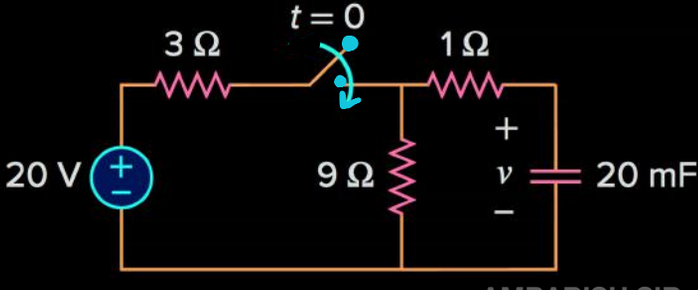
Considering Yellow line as base, $V_{eff} = V + V_0$.

CAPACITOR CHARGING IN A NON SERIES CIRCUIT:

If we have a circuit with single capacitor and multiple batteries and resistors, then to find charge on capacitor and its current, we can always thevenize the capacitor to form a simple series effective circuit and then backwork to find various currents in original circuit.

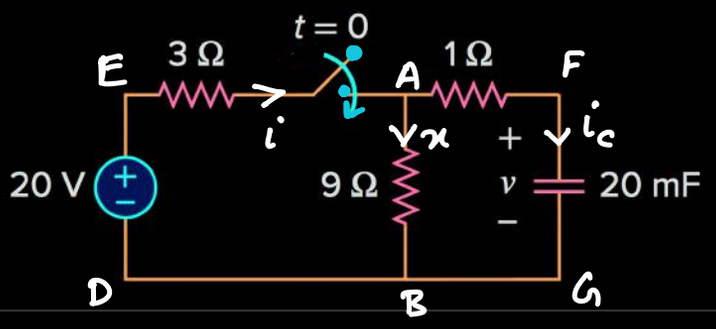


EXAMPLE: In adjacent circuit, switch was closed at $t = 0$. Find:
(a) Steady state charge on capacitor
(b) Time constant of capacitor charging
(c) Charge on capacitor as a function of time.



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(d) Current through battery as a function of time.



Solution: $\mathcal{E}_H = 15\,V$, $R_H = 13/4\,\Omega$

⇒ (a) $q_\infty = C\mathcal{E}_H = 300\,mC$ Ans.

(b) $\tau = R_H C = 65\,ms$.

(c) $q = q_\infty (1 - e^{-t/\tau})$

$\Rightarrow q = 300\,(1 - e^{-1000t/65})\,mC$ Ans.

(d) $i_c = dq/dt = \frac{\mathcal{E}_H}{R} e^{-t/\tau} = \frac{60}{13} e^{-1000t/65}$ A

Now applying KVL along EAFGBDE: $3i + i_c + q/C - 20 = 0$

$$\Rightarrow i = (20 - i_c - q/C) \times \frac{1}{3} = \frac{1}{3}\left[20 - \frac{60}{13} e^{-1000t/65} - 15\,(1 - e^{-1000t/65})\right]$$

$$\Rightarrow i = \frac{1}{3}\left[5 + \frac{135}{13} e^{-1000t/65}\right] A \text{ Ans.}$$

Alternatively we could find current through the battery as:

$$i = i_0 + (i_\infty - i_0)\left[1 - e^{-t/\tau}\right] = i_\infty - (i_\infty - i_0)\, e^{-t/\tau}$$

For finding i_∞, capacitor can be replaced by open circuit and for finding i_0, capacitor can be replaced by short circuit.

For our circuit, $i_0 = \frac{200}{39}$ A, $i_\infty = \frac{5}{3}$ A, thus the required current $i = \frac{5}{3} + \frac{135}{39} e^{-1000t/65} = \frac{1}{3}\left[5 + \frac{135}{13} e^{-1000t/65}\right]$

This is same answers as before.

(Here we have used the fact that at any instant capacitor behaves like a battery whose EMF changes by a factor of $e^{-t/\tau}$ while other batteries in circuit are constant, thus current in any branch must be of the form $i = a + b\,e^{-t/\tau}$ and should satisfy that @ $t=0$, $i = i_0$ and at $t=\infty$, $i = i_\infty$, thus,

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$a = i_\infty$ and $b = (i_0 - i_\infty)$, thus:

→ Current through any branch is of the form $\boxed{i = i_\infty + (i_0 - i_\infty)e^{-t/\tau}}$

→ Thus if we know time constant of any circuit, we only need to find i_0 and i_∞ through any branch to find current as a function of time.

→ A similar formula would also hold for voltages and charges i.e.:

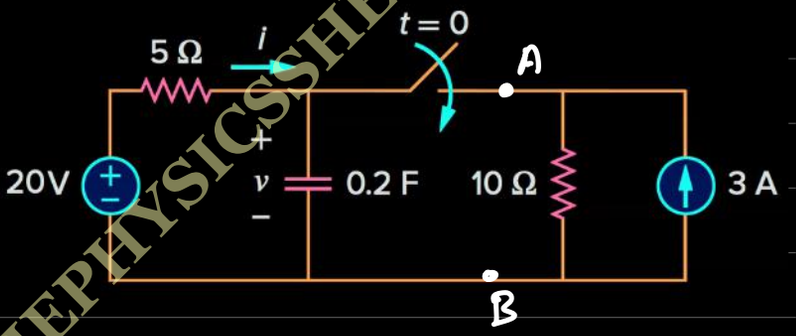
$$\boxed{V = V_\infty + (V_0 - V_\infty)e^{-t/\tau}} \quad \text{and} \quad \boxed{q = q_\infty + (q_0 - q_\infty)e^{-t/\tau}}$$

EXAMPLE: In the shown circuit, initially switch was open since a long time and then it is closed at $t=0$ find:

(a) Time constant after closing the switch.

(b) $v(t)$ across capacitor at general time

(c) $i(t)$ through 5Ω resistor.



Solution: At $t=0$, across capacitor, $V_0 = 20$ V

Now using Millman's theorem, $\mathcal{E}_{eff} = \left(\frac{20/5 + 0/10 + 3}{1/5 + 1/10 + 1/\infty}\right) = \frac{70}{3}$ Volts

Also $R_{eff} = \left[\frac{1}{5} + \frac{1}{10}\right]^{-1} = 10/3\ \Omega$

$\Rightarrow \tau = C \times R_{eff} = 0.2 \times 10/3 = 2/3$ sec. Ans (a)

Now $V_\infty = \mathcal{E}_{eff} = 70/3$ Volt thus across capacitor:

$$v = V_\infty + (V_0 - V_\infty)e^{-t/\tau} = \frac{70}{3} - \frac{10}{3}e^{-3t/2} \text{ Ans}$$

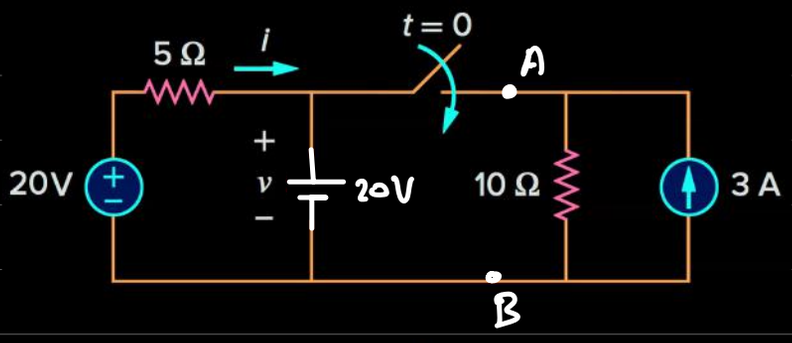
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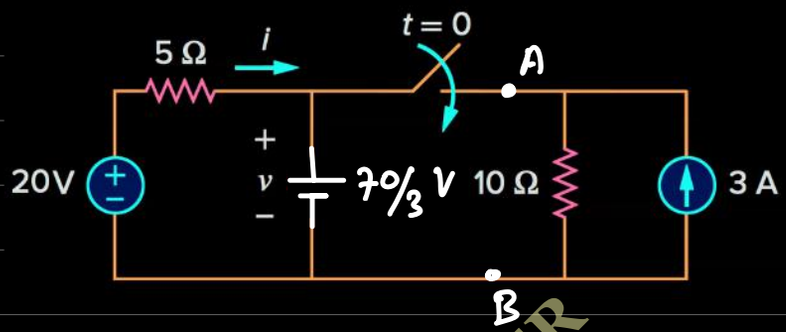
Now at $t = 0^+$, $i_0 = 0$
(See adjacent figure).
also at $t = \infty$, $i = \frac{(20 - 70/3)}{5}$

$\Rightarrow$ $i_\infty = -2/3$ Ampere

Now $i = i_\infty + (i_0 - i_\infty) e^{-t/\tau}$

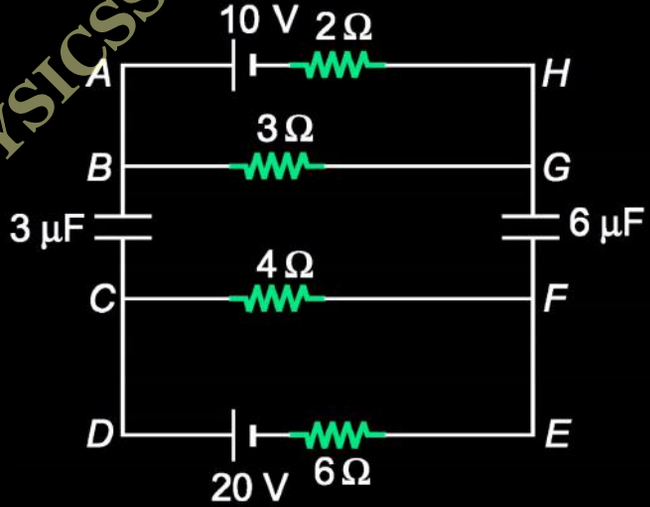
Thus $i = \frac{-2}{3} + \frac{2}{3} e^{-2t/3}$



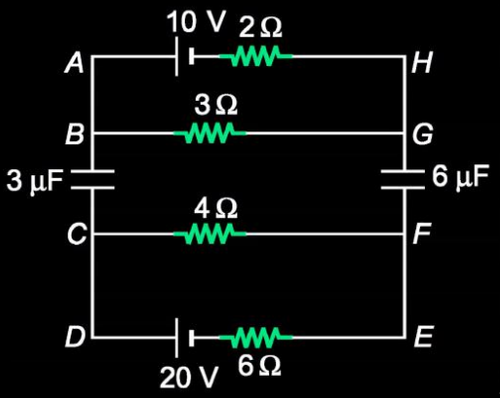


Circuits With Multiple Capacitors In Steady State:

In a circuit with multiple capacitors, in general it will not be possible to define a unique time constant (except for simple circuits which can be simplified in to series or parallel equivalents), however we may still be able to find final charges using the fact that at steady state, capacitors behave like open circuit, thus currents in other branches can be readily found, by replacing capacitors by open circuits (now the circuit is standard battery and resistors circuit). After figuring out currents, we can figure out various potential differences to work out the individual charges, using KVL or direct observations.

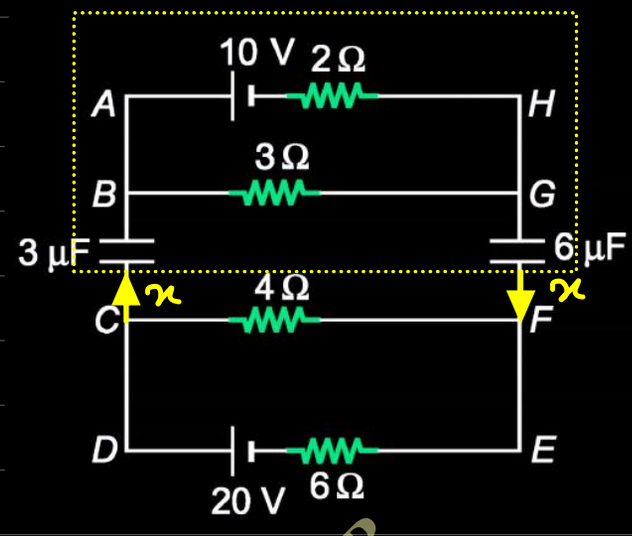


Example: (a) Find the steady state charges on the two capacitors in the adjacent circuit. (b) Is the time constant of the circuit defined (c) Can you find charges as a function of time assuming



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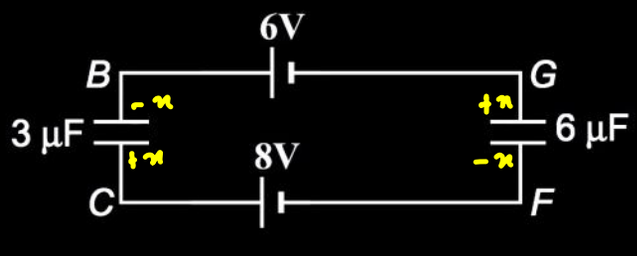
that both batteries were turned on simultaneously at $t=0$.



Solution: Consider the system boundary drawn in yellow. The net charge entering this boundary must be equal to that exiting this boundary, thus charges on both capacitors must be equal at all times. Now in steady state both capacitors behave as open circuits. Thus current in 3Ω resistor, $I_3 = 10/5$ i.e 2A and $V_{BG} = 2 \times 3 = 6$ Volts,

Similarly across 4Ω resistor $V_{CD} = 20 \times \frac{4}{10} = 8$ V.

Now using compensation theorem we can show the following circuit in steady state:



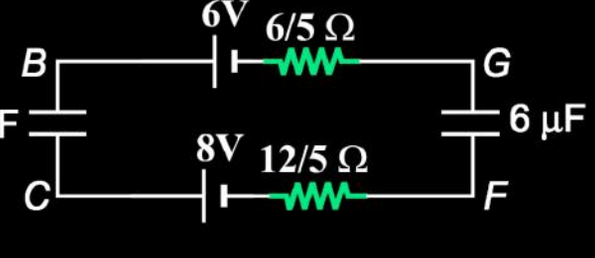
Now this is like series combination of 3μF and 6μF against an effective battery of 2V, thus, final charges:

$x = \left(\frac{6 \times 3}{6+3}\right) \times 2\ \mu C = 4\ \mu C$ ans (a)

(b) For finding time constant we shall thevenize ABGH between points B and G and also thevenize loop FEDC between points F and C let us call corresponding thevenin EMFs and resistances as $\mathcal{E}_1, R_1, \mathcal{E}_2$ and R_2 respectively. Thus:

$\mathcal{E}_1 = 6$ V, $R_1 = 3 \times 2/(3+2) = 6/5\ \Omega$, $\mathcal{E}_2 = 8$ V, $R_2 = 4 \times 6/(4+6) = 12/5\ \Omega$

Now our net equivalent circuit is as shown in adjacent figure. Now:



$C_{eff} = \frac{3 \times 6}{3+6} = 2\ \mu F$ $R_{eff} = \frac{6}{5} + \frac{12}{5} = \frac{18}{5}\ \Omega$

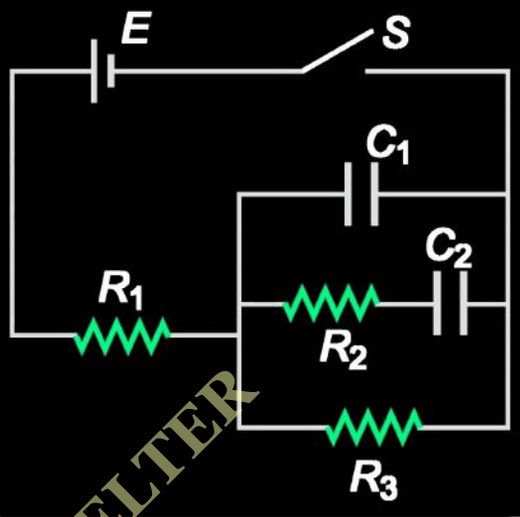
$\tau_{eff} = C_{eff} R_{eff} = 36/5\ \mu$ Sec ans (b)

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Now charge on capacitors, $q = C_{eff}\mathcal{E}_{eff}(1 - e^{-t/\tau})$, thus:

$$q = 4\left(1 - e^{-5\times10^6 t/36}\right)$$ Ans (c).

EXAMPLE: For the circuit shown in adjacent figure find (a) Current from battery immediately after closing switch (b) Battery current in steady state. (c) Potential drop across capacitors in steady state.



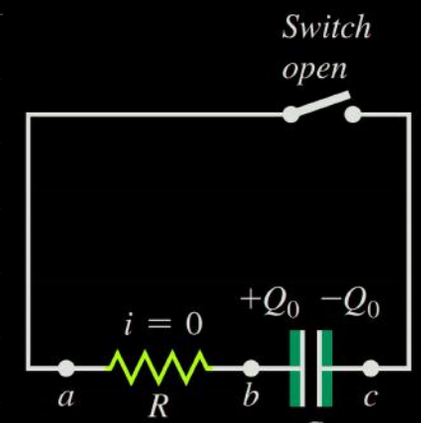
Solution: At $t=0$, C_1 and C_2 are like short circuits. Thus effective resistance is only R_1 Thus Battery current at $t=0$ $i_0 = \mathcal{E}/R_1$ Ans (a). Now in steady state, C_1 and C_2 are open circuit, thus effective resistance $= R_1 + R_3$ thus from battery steady state current $i_\infty = \mathcal{E}/(R_1+R_3)$ Ans (b). Now in steady state R_1 and R_3 are in series thus potential drop across capacitor: $V_3 = \dfrac{R_3}{R_3+R_1}\times E$ Ans.

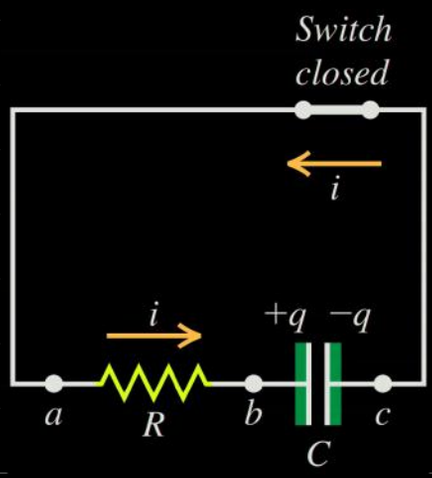
RC DISCHARGING CIRCUITS:

Consider the circuits shown in the adjacent figure. Here the capacitor is initially charged and the switch is open. Now when switch is closed, the capacitor will start getting discharged through the resistor R and we wish to know the charge in the capacitor as a function of time. In principle this problem is similar to that of charging a precharged capacitor except that here $\mathcal{E}_R = 0$ and $R_R = R$ and we can directly write solutions for charge, potential and current:

$$q = q_\infty + (q_0 - q_\infty)e^{-t/\tau}, \text{ thus:}$$

$$q = 0 + (Q_0 - 0)e^{-t/RC} \text{ i.e } \boxed{Q = Q_0 e^{-t/RC}}$$





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Here Q_∞ is zero because finally capacitor will get fully discharged.
Now $i_0 = -Q_0/RC$ (negative sign because current flows opposite to shown direction. Now:

$$i = i_\infty + (i_0 - i_\infty)e^{-t/\tau} = 0 + (-Q_0/RC - 0)e^{-t/RC} = \frac{-Q_0}{RC}e^{-t/\tau}$$

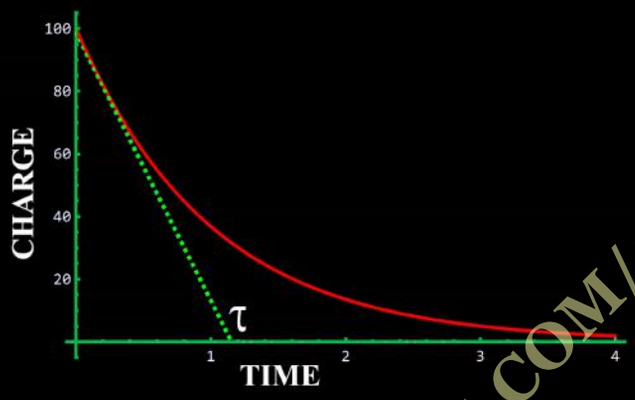
Now if we just want magnitude of discharging current we can write:

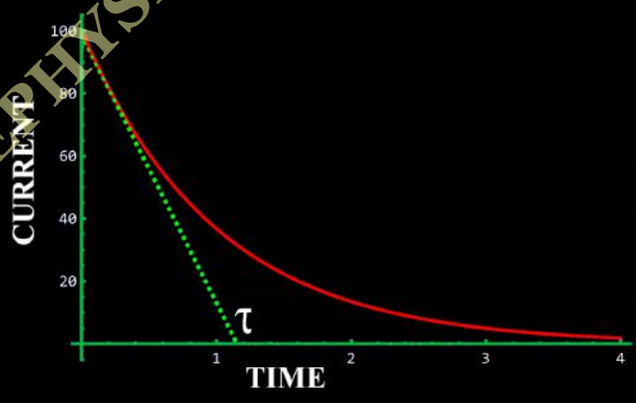
$$\boxed{|i| = |i_0|e^{-t/\tau}} \quad \text{where } i_0 = V_0/R = Q_0/RC$$

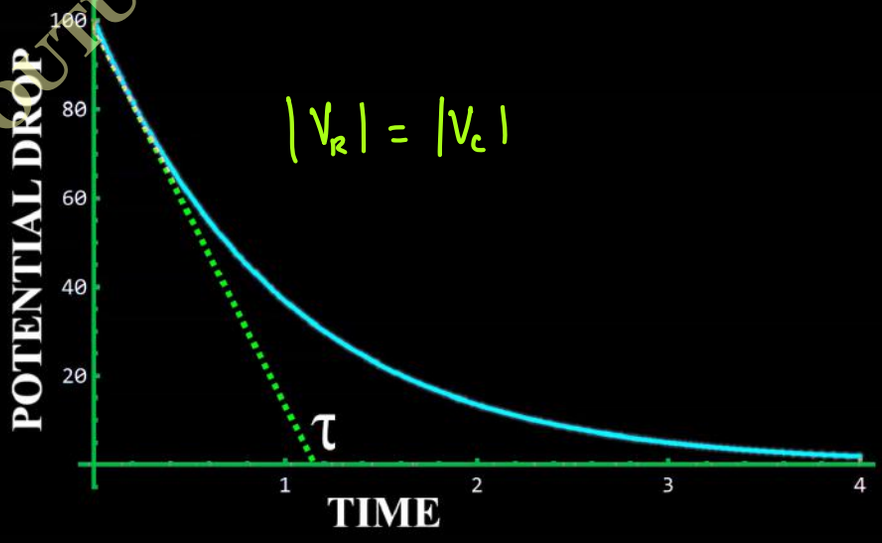
Similarly we can write potential difference as $V = V_\infty + (V_0 - V_\infty)e^{-t/\tau}$ thus:

$$\boxed{V = V_0 e^{-t/\tau}} \quad \text{where } V_0 = Q_0/C$$

Charge Voltage And Current Plots For Discharging Circuits:







$$|V_R| = |V_C|$$

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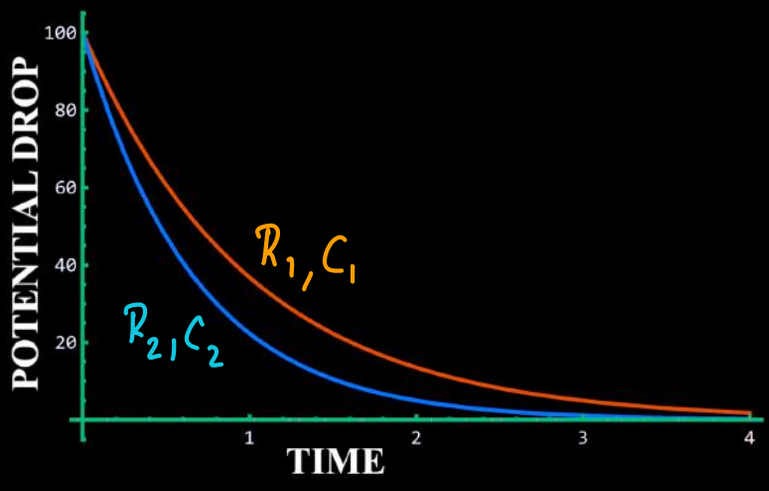
Example: Adjacent figure shows discharging graphs of two capacitors having same initial charges. Then we can conclude that:

(a) $R_1C_1 > R_2C_2$

(b) $R_1 > R_2$

(c) $C_1 > C_2$

(d) $C_1 = C_2$



Since initial potential drops are same, $\therefore Q_1/C_1 = Q_2/C_2$ but it is given that $Q_1 = Q_2 \Rightarrow C_1 = C_2$ (d) also tangent intercept for C_2 is smaller thus $R_2C_2 < R_1C_1$ (a) also this implies $R_1 > R_2$ (b) ($\because C_1 = C_2$). Thus correct answers are (a), (b) and (d).

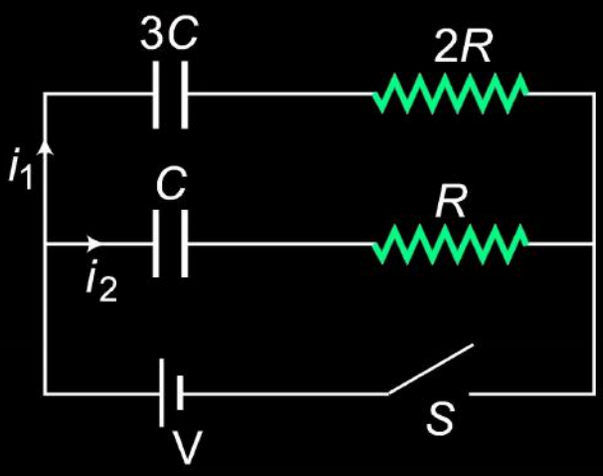
Think: What would be your answer to previous problem if it were given that initial currents in circuits are equal instead of charges?

Solution: If $i_1 = i_2 \Rightarrow \frac{Q_1}{R_1C_1} = \frac{Q_2}{R_2C_2}$ ① but $R_1C_1 > R_2C_2$ (a) thus $Q_1 > Q_2$, Also initial potential differences are equal thus $\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$ ② Thus $R_1 = R_2$ ($\because$ ① & ②), Now $\frac{C_2}{C_1} = \frac{Q_2}{Q_1} \cdot \frac{R_1}{R_2} < 1 \Rightarrow C_2 < C_1$

thus (c) is also correct. Thus correct options would be (a) and (c)

Parallel Branches Across Ideal Battery:

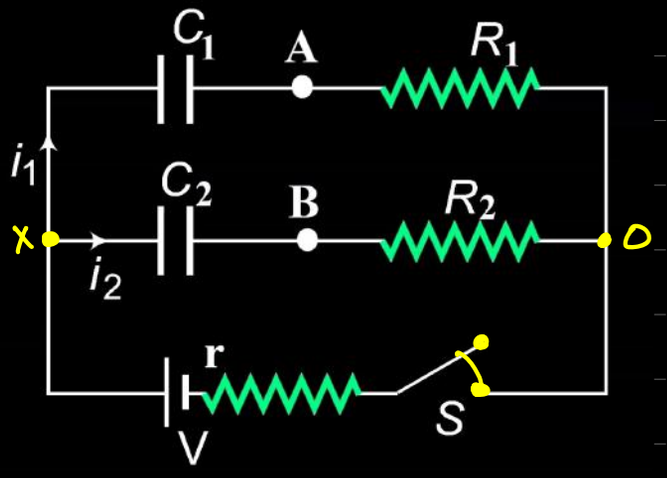
If there are several RC branches connected across an ideal battery (zero resistance) in parallel then each branch carries charge according to its own time constant. For instance in shown adjacent circuit time constant of upper branch is 6RC and that of lower RC branch is RC. Thus $i_1 = \frac{V}{2R}e^{-t/6RC}$ and $i_2 = \frac{V}{R}e^{-t/RC}$



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Parallel Branches With Equal RC Products Across Real Battery:

When we have parallel RC Branches as shown in adjacent circuit such that $R_1C_1 = R_2C_2$ then we can prove that A and B will be equipotential (and hence can be fused together to form simple series parallel circuit). This can be shown as follows:



Applying Junction law at node A:

$$\frac{V_A}{R_1} + C_1 \frac{d}{dt}(V_A - X) = 0$$

$$\Rightarrow \quad V_A + R_1C_1\dot{V}_A = R_1C_1\dot{X}(t)$$

Now let $R_1C_1 = R_2C_2 = T$, then D.E. governing V_A is:

$$V_A + T\dot{V}_A = T\dot{X}(t) \quad \text{and} \quad V_A(0) = X(0) \qquad (1)$$

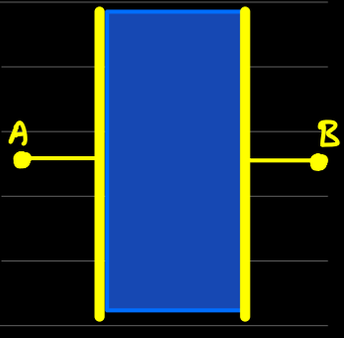
Similarly $V_B + T\dot{V}_B = T\dot{X}(t)$ and $V_B(0) = X(0)$ (2)

Thus both V_A and V_B are both governed by same first order differential equation with same initial conditions thus they have same solutions i.e $V_A = V_B$ at all times, Now if we we fuse A & B, $C_1 \| C_2$ and $R_1 \| R_2$, we can easily solve for currents / charges in this circuit and divide individually in appropriate ratios.

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Example: Irodov 3.160:

The gap between the plates of a parallel-plate capacitor is filled with glass of resistivity $\rho = 100$ GΩ·m. The capacitance of the capacitor equals $C = 4.0$ nF. Find the leakage current of the capacitor when a voltage $V = 2.0$ kV is applied to it. Dielectric constant of glass is k.



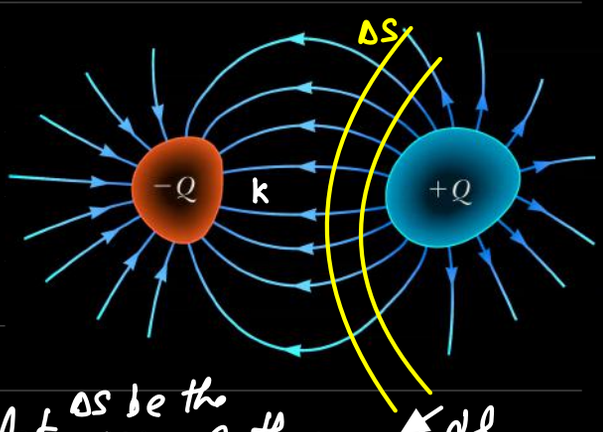
Solution: $J = E/\rho \Rightarrow I = EA/\rho = VA/\rho d$

Also $C = \epsilon_0 kA/d \Rightarrow A/d = C/\epsilon_0 k$

$$\Rightarrow \quad I = \frac{V}{\rho} \cdot \frac{C}{\epsilon_0 k} \quad \text{Ans}$$

Example: Irodov 3.161:

Two conductors of arbitrary shape are embedded into an infinite homogeneous poorly conducting medium with resistivity ρ and dielectric constant k. Find the value of a product RC for this system, where R is the resistance of the medium between the conductors, and C is the mutual capacitance of the conductors in the presence of the medium.



Solution: Consider two nearby equipotentials in a region of thickness dl and let ΔS be the area of these equipotentials. Now for this small volume, elemental resistance:

$$dR = \frac{\rho\, dl}{\Delta S}$$ and elemental reciprocal capacitance:

$$d\left(\frac{1}{C}\right) = \frac{dl}{\epsilon_0 k \Delta S} \quad (\because C = \epsilon_0 A/d \text{ for } \| \text{ plate capacitors})$$

Thus:

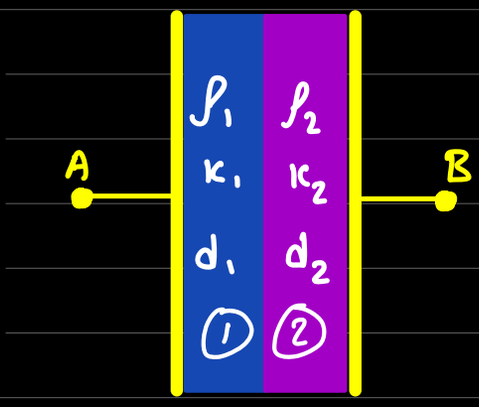
$$dR = d\left(\frac{1}{C}\right) \times \rho \epsilon_0 k$$ now integrating this from one conductor to the other:

$$R = \frac{1}{C} \rho \epsilon_0 k \Rightarrow RC = \rho \epsilon_0 k$$

Thus for arbitrary conductor capacitor time constant $\boxed{\tau = \rho \epsilon_0 k = \rho \epsilon}$

Example: Irodov: 3.166:

The gap between the plates of a parallel-plate capacitor is filled up with two dielectric layers 1 and 2 with thicknesses d_1 and d_2, dielectric constants k_1 and k_2, and resistivities ρ_1 and ρ_2. A dc voltage V is applied to the capacitor, with electric field directed from layer 1 to layer 2. Find σ, the surface density of extraneous charges at the boundary between the dielectric layers, and the condition under which $\sigma = 0$.



Solution: Effective resistance: $R = \frac{\rho_1 d_1}{A} + \frac{\rho_2 d_2}{A}$

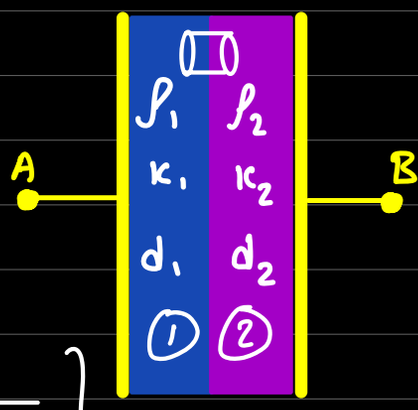
$\Rightarrow$ Current $I = \frac{V}{R} = \frac{VA}{\rho_1 d_1 + \rho_2 d_2}$ $\Rightarrow J = \frac{I}{A} = \frac{V}{\rho_1 d_1 + \rho_2 d_2}$

Also $J = E_1/\rho_1 = E_2/\rho_2$ (E_1 & E_2 are field in regions ① and ②)

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$$\Rightarrow \quad E_1 = \rho_1 V\left[\frac{1}{\rho_1 d_1 + \rho_2 d_2}\right] \text{ and } E_2 = \rho_2 V\left[\frac{1}{\rho_1 d_1 + \rho_2 d_2}\right]$$

Now imagine a Gaussian surface near dielectric interface,



Using Gauss law for dielectrics:

$$k_1 E_1 S - k_2 E_2 S = \frac{\sigma \times S}{\epsilon_0}$$

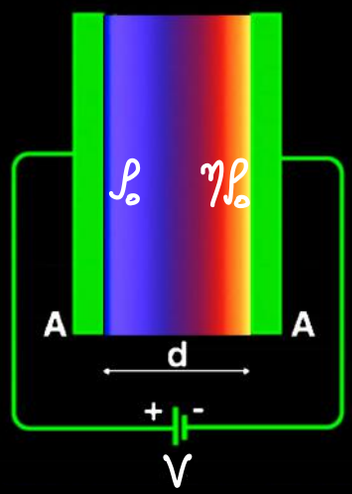
$$\Rightarrow \quad \sigma = \epsilon_0 [k_1 E_1 - k_2 E_2]$$

$$\Rightarrow \quad \sigma = \epsilon_0 \rho_1 V k_1 \left[\frac{1}{\rho_1 d_1 + \rho_2 d_2}\right] - \epsilon_0 \rho_2 V k_2 \left[\frac{1}{\rho_1 d_1 + \rho_2 d_2}\right]$$

$$\Rightarrow \sigma = \epsilon_0 V (\rho_1 k_1 - \rho_2 k_2)\left[\frac{1}{\rho_1 d_1 + \rho_2 d_2}\right] \text{ Ans.}$$

EXAMPLE: IRODOV: 3.168:

The space between the plates of a parallel-plate capacitor is filled up with inhomogeneous poorly conducting medium whose resistivity varies linearly in the direction perpendicular to the plates. The ratio of the maximum value of resistivity to the minimum one is equal to η. The gap width equals d. Find the volume density of the charge in the gap if a voltage V is applied to the capacitor. k is assumed to be 1 everywhere.



Solution: $\rho(x) = \rho_0 + \frac{(\eta-1)\rho_0 x}{d} = \rho_0\left[1 + \frac{(\eta-1)x}{d}\right]$

Now $R = \int \frac{\rho dx}{A}$ and $I = \frac{V}{R} \Rightarrow J = \frac{V}{AR} = \frac{E}{\rho}$

$\Rightarrow E = \rho\frac{V}{AR} = \frac{\rho V}{\int_0^d \rho dx}$, Now $\vec{\nabla}\cdot\vec{E} = \rho_c/\epsilon_0$ ($\rho_c \rightarrow$ charge density)

$$\Rightarrow \rho_c = \epsilon_0 \frac{d}{dx}\left[\frac{\rho V}{\int_0^d \rho dx}\right] = \frac{\epsilon_0 V}{\int_0^d \rho dx}\left(\frac{d\rho}{dx}\right)$$

Now $d\rho/dx = \rho_0(\eta-1)/d$ and $\int_0^d \rho dx = \rho_0\left[x + \frac{(\eta-1)x^2}{2d}\right]_0^d = \frac{d}{2}(1+\eta)\rho_0$

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Now substituting these, $\rho_c = \varepsilon_0 V \cdot \frac{\rho_0(\eta-1)}{d} \cdot \frac{2}{\rho_0(1+\eta)d}$

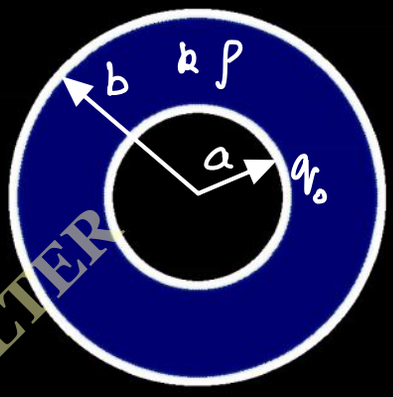
$\Rightarrow \rho_c = \frac{2\varepsilon_0 V}{d^2}\left(\frac{\eta-1}{\eta+1}\right)$ Ans.

EXAMPLE: IRODOV: 3.203:

The radii of a spherical capacitor electrodes are equal to a and b with a<b. The interelectrode space is filled with homogeneous substance of dielectric constant k and resistivity ρ. Initially the capacitor is not charged. At the moment t = 0 the internal electrode gets a charge q_0. Find:

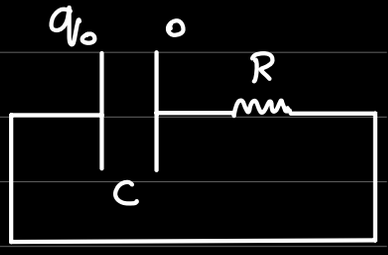
(a) the time variation of the charge on the internal electrode;

(b) the amount of heat generated during the spreading the charge.



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Solution: Equivalent circuit is shown in the figure.



Recall that time constant for such an arrangement is $\rho\varepsilon_0 k$ as proved in a previous example.

Also we can break the capacitor in to a battery of Voltage $V_0 = \frac{1}{4\pi\varepsilon_0 k}\left[\frac{q_0}{a} - \frac{q_0}{b}\right]$ and an empty capacitor.

$V_0 = \frac{q_0}{C}$ where $C = \frac{4\pi\varepsilon_0 k\, ab}{b-a}$

Now charge flown at a general time $x = CV_0(1-e^{-t/\tau})$

$= q_0(1-e^{-t/\tau})$

Thus net charge on inner electrode, $q = q_0 - x = q_0 e^{-t/\tau}$

Thus:

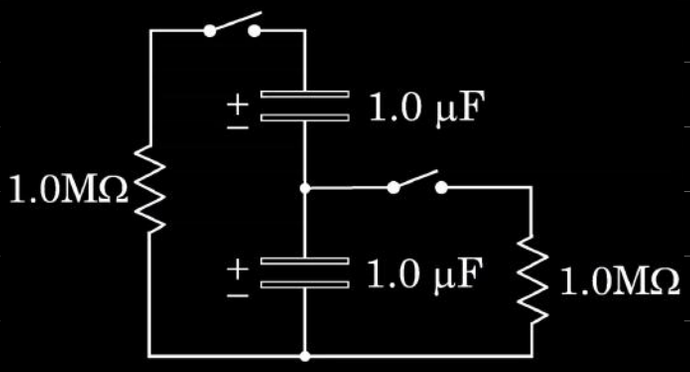
$q = q_0 e^{-t/\rho\varepsilon_0 k}$ Ans. (Finally all charge will flow outwards).

Now heat liberated is same as the energy lost from dielectric region (which is same as if inner electrode had q_0 and outer $-q_0$)

Thus heat liberated : $\Delta H = q_0^2/2C$

$\Rightarrow \Delta H = q_0^2 \Big/ 2\left(\frac{4\pi\varepsilon_0 abk}{b-a}\right) = \frac{q_0^2(b-a)}{8\pi\varepsilon_0 kab}$ Ans.

RC Circuits With Multiple Capacitors:



Transient response of RC circuits with arbitrary number of capacitors in arbitrary configurations can lead to systems of complicated systems of simultaneous linear differential equations which is beyond scope of JEE. However, pathfinder has one such problem with some special values. Here we shall provide a general procedure for solving such circuits through a simple example. Before we get to the procedure we need to define some notation:

Differential Operator (D): Differential operator D simply means differentiation wrt time, thus Dx means $\frac{dx}{dt}$. D can also be used to form polynomial operators.
Thus $(D^2+2D+1)x$ means $\frac{d^2x}{dt^2}+2\frac{dx}{dt}+x$

→ Operator D in many ways behaves just like an algebraic variables for example:

$$(D+1)^2x = (D^2+2D+1)x = \frac{d^2x}{dt^2}+2\frac{dx}{dt}+x$$

This can be readily verified as follows:

$$(D+1)^2x = (D+1)[(D+1)(x)] = (D+1)\left[\frac{dx}{dt}+x\right]$$

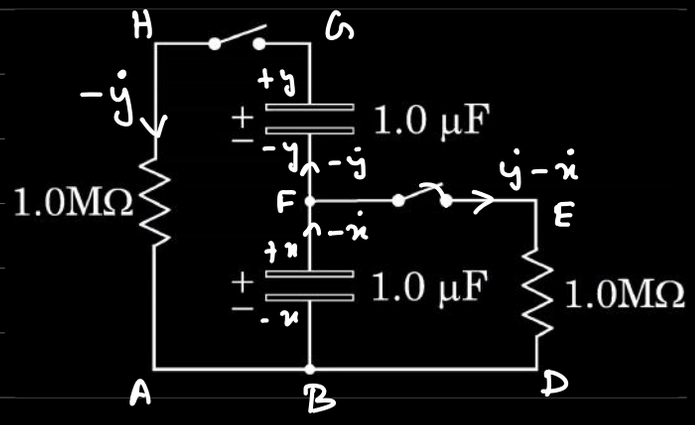
$$= \frac{d^2x}{dt^2}+\frac{dx}{dt}+\frac{dx}{dt}+x = \frac{d^2x}{dt^2}+2\frac{dx}{dt}+x \quad \text{QED.}$$

→ If any equation is identically true, same Differential polynomial operator can be applied on both sides of the equation and it will still yied a valid differential equation.

→ Equations can be operated upon by suitable Differential polynomial operators to eliminate suitable variables and obtain differential equations in a single variable.

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EXAMPLE: In the adjacent circuit, both capacitors are charged to a potential of $V_0 = 1$ Volts. (a) Develop linear differential equations to solve for instantaneous charges on the capacitors (y and x are values of instantaneous charges). (b) How will you solve these differential equations. (x and y are in microcoulombs)



Solution: Consider the loop ADGHA, Applying KVL

$$-x - y - \dot{y} = 0 \quad \Rightarrow \quad (D+1)y + x = 0 \quad (1)$$

Similarly for loop FEDBF:

$$\dot{y} - \dot{x} - x = 0 \quad \Rightarrow \quad Dy - (D+1)x = 0 \quad (2)$$

Now $(D+1)(1) + (2)$ yields: $\left[(D+1)^2 + D\right]y = 0$

$$\Rightarrow \left(D^2 + 3D + 1\right)y = 0 \quad \text{i.e} \quad \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + y = 0$$

Recall this is standard differential equation of the kind we came across while studying damped oscillations. For solving this, we make allied quadratic equation (or characteristic equation), i.e:

$$\lambda^2 + 3\lambda + 1 = 0 \quad \Rightarrow \quad \lambda = \left(\frac{-3 \pm \sqrt{5}}{2}\right)$$

Now general solution for y is $Ae^{\lambda_1 t} + Be^{\lambda_2 t}$

The constants A and B can be evaluated using the initial conditions i.e $y(0) = 1$ and $\dot{y}(0) = -2$ $\left(\because y(0) = CV_0,\ \dot{y}(0) = \frac{-2V_0}{R}\right)$

Using these initial conditions, $y(t)$ can be found as:

$$y = \frac{5+\sqrt{5}}{10}\, e^{-\left(\frac{3+\sqrt{5}}{2}\right)t} + \left(\frac{5-\sqrt{5}}{10}\right) e^{\left(\frac{-3+\sqrt{5}}{2}\right)t}$$

Now $x = -(D+1)y$, Alternatively $D(1) - (D+1)(2)$ yields:

$$\left[D + (D+1)^2\right]x = 0 \quad \text{i.e} \quad \left(D^2 + 3D + 1\right)x = 0$$

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or $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 1 = 0$ thus $x = Ce^{-\lambda_1 t} + De^{-\lambda_2 t}$

where $\lambda = \frac{-3 \pm \sqrt{5}}{2}$ also $x(0) = 1$ and $\dot{x}(0) = -$current through resistors at $t=0$, i.e $-(1+2) = -3$. Now using $x(0)$ and $\dot{x}(0)$ we can evaluate C and D. Evaluating we get:

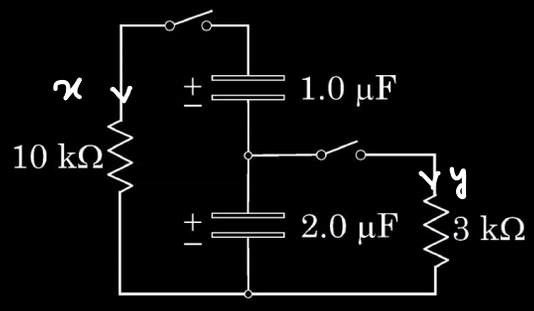
$$x = \frac{1}{10}\left[(5+3\sqrt{5})e^{-\left(\frac{3+\sqrt{5}}{2}\right)t} + (5-3\sqrt{5})e^{\left(\frac{\sqrt{5}-3}{2}\right)t}\right] \text{ Ans.}$$

→ If $y(0)$ and $\dot{y}(0)$ are both scaled by some factor η one can readily see that a solution scaling up A and B by η will satisfy the new initial conditions i.e new solution will be simply,

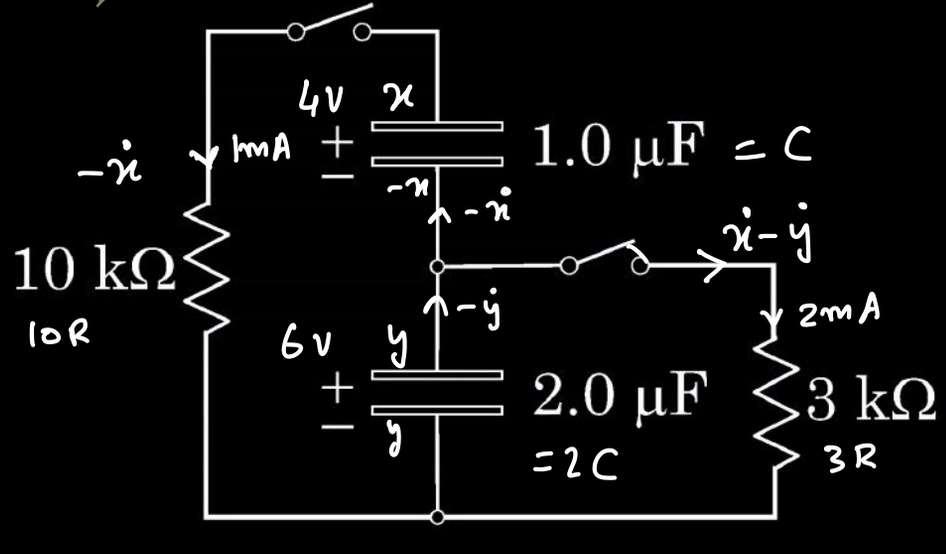
$$\eta\left[Ae^{-\lambda_1 t} + Be^{-\lambda_2 t}\right].$$

EXAMPLE : PATHFINDER : BUILD: 52 :

The circuit shown consists of two capacitors of capacitances 1.0 μF and 2.0 μF, two resistors of resistances 10 kΩ and 3 kΩ and two switches. Initially the switches are open and the capacitors are charged to potentials 4.0 V and 6.0 V respectively according to the polarities shown. If both the switches are closed simultaneously, what are amounts of heat dissipated in the 10 kΩ and the 3 kΩ resistors over a time large enough for complete discharge of both the capacitors?



Solution: Let x and y denote instantaneous charges on the two capacitors then $x(0) = 4\mu C$, $y(0) = 12\mu C$, $\dot{x}(0) = -1mA$, $\dot{y}(0) = -3mA$



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Observe that initial conditions of y are scaled from initial conditions of x by a factor of 3.

Now applying KVL to the left loop:

$$-y/2C - x/C - 10iR = 0$$

$$\Rightarrow \left(10RD + \frac{1}{C}\right)x + \frac{1}{2C}y = 0 \quad (1)$$

Similarly for right loop

$$-y/2C + (i - j) \times 3R = 0$$

$$\Rightarrow 3RD\,x - \left(3RD + \frac{1}{2C}\right)y = 0 \quad (2)$$

Eliminating y from (1) and (2), we get:

$$\left(60R^2D^2 + \frac{19R}{C}D + \frac{1}{C^2}\right)x = 0 \quad (3)$$

Similarly eliminating x we get:

$$\left(60R^2D^2 + \frac{19R}{C}D + \frac{1}{C^2}\right)y = 0 \quad (4)$$

Now observe that (3) and (4) are the same differential equations and initial conditions for x and y are just scaled versions of each other, Thus the solutions for x and y are just scaled versions of each other, $y = 3x$ to be precise. Thus ratio of charges on capacitors and hence ratio of currents in the resistors maintain a constant ratio of 1:2 (10R:3R) that is the initial ratio of currents). Thus ratio of heat released:

$$H_{10R} : H_{3R} = 1^2 \times 10 : 2^2 \times 3 \text{ i.e. } 10 : 12 \text{ i.e } 5 : 6$$

Now Total heat released = Total energy stored in capacitors i.e.

$$\frac{1}{2} \times 1\mu F \times 4^2 + \frac{1}{2} \times 2\mu F \times 6^2 = 44\mu J$$

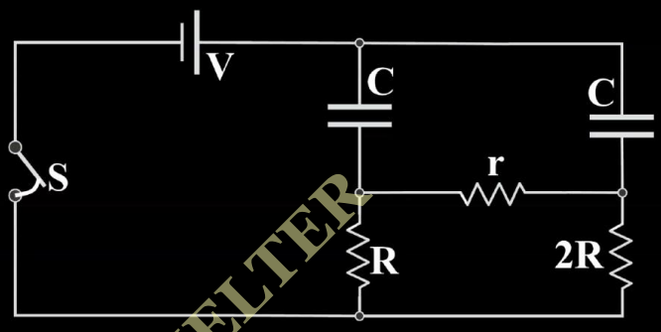
Thus heat generated in $10R : \frac{5}{11} \times 44\mu J = 20\mu J$ and in $3R = 24\mu J$ Ans.

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→ It is not by chance that both variables have same differential equations, it can be shown using eigen value/eigen vector theory that this will always be the case, however it is way beyond scope of our purpose here.

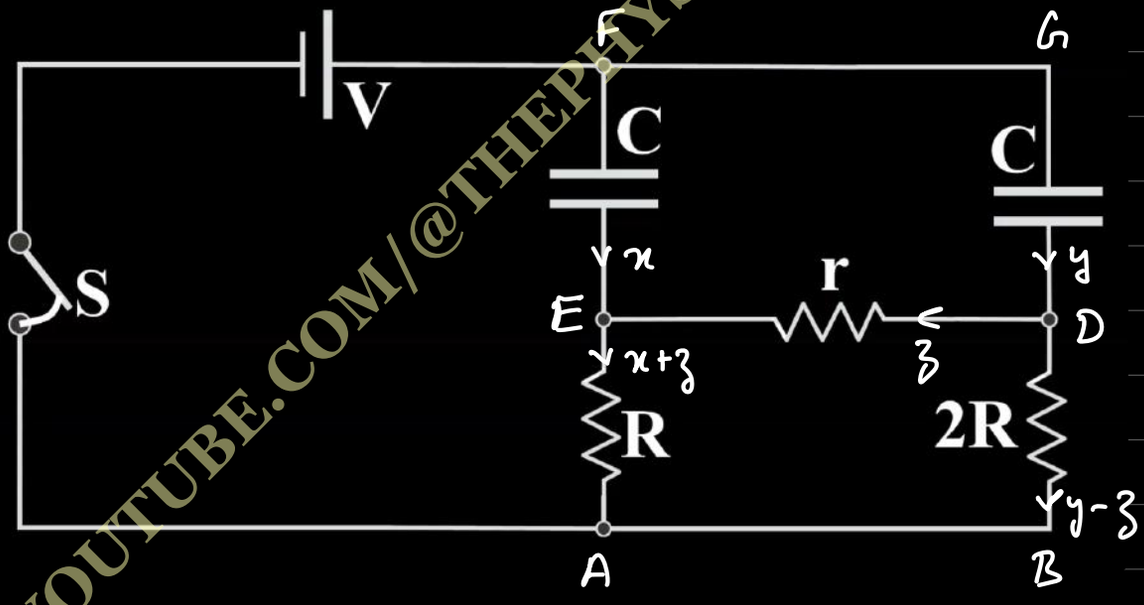
→ We could solve this question with relatively less effort due to initial conditions being simply scaled versions of each other, otherwise this question would require a lot of dull work.

EXAMPLE: In the shown adjacent circuit, the switch was closed at $t=0$, find the total charge flown through $\mathcal{R}$ until steady state is achieved.



Solution: Let x, y and z be instantaneous currents as shown.

Now find charges on capacitor $Q = CV = \int x\,dt = \int y\,dt$



Now applying kVL to loop ABDE: $\mathcal{R}z + (x+z)R - (y-z)\times 2R = 0$

$$\Rightarrow z = \frac{R(2y-x)}{\mathcal{R}+3R}$$ Now total charge passing through $\mathcal{R} = \int z\,dt$.

Integrating above equation, total charge flown along z:

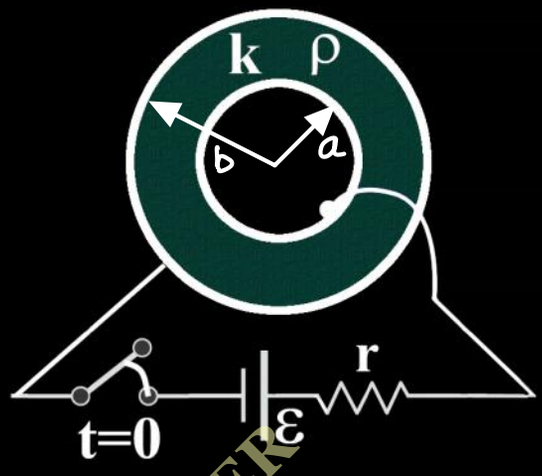
$$Z = \frac{R(2Q-Q)}{\mathcal{R}+3R} = \frac{CVR}{\mathcal{R}+3R}$$

→ The key idea here was to use kVL along with knowing

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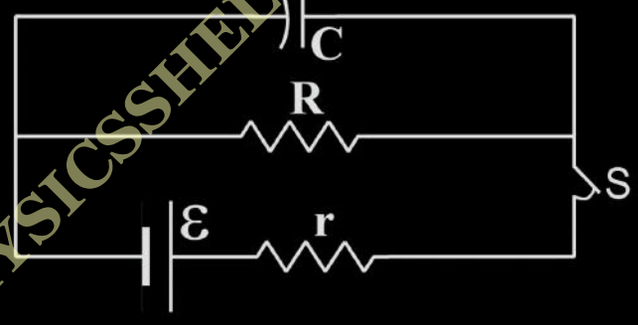
the integrals of some currents before hand. This technique can be used in other similar problems also.

EXAMPLE: Consider a system of concentric spherical shells capacitor as shown in the figure. The dielectric has a resistivity ρ. Find:
(a) Time constant of the circuit
(b) Steady state current through battery
(c) Maximum charge on the capacitor.



Solution: The potential difference across inner and outer electrode of capacitor is same as that across the resistance between these electrodes. Now:

$$C = \frac{4\pi\varepsilon_0 k\, ab}{b-a}, \quad R = \frac{\rho(b-a)}{4\pi ab}$$



Now $\mathcal{E}_{th} = \frac{\mathcal{E}R}{R+r}$, $r_{th} = \frac{Rr}{R+r}$, thus:

Time Constant: $\tau = r_{th} C = \frac{Rr}{R+r} C$ Ans. (a)

Steady State Current: $\mathcal{E}/R+r$ Ans (b)

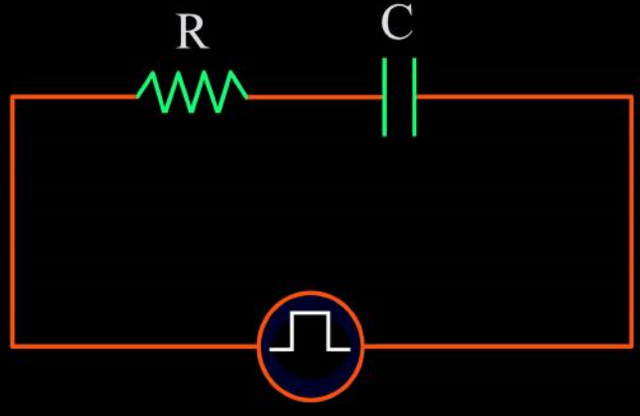
Maximum charge on capacitor: $\frac{\mathcal{E}RC}{R+r}$ Ans (c)

RC-CIRCUIT WITH SQUARE TOOTH WAVE (STEADY STATE)

Consider an RC circuit subjected to square tooth wave i.e:

$\mathcal{E}(t) = V$ if $[t/\lambda]$ is even

$\mathcal{E}(t) = 0$ if $[t/\lambda]$ is odd.

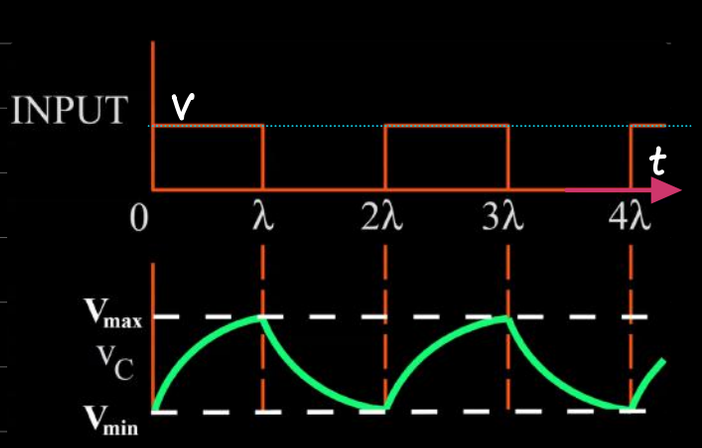


The initial response to this EMF is not periodic, however after a long time the potential difference across the capacitor becomes periodic, oscillating between

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certain maximum and minimum value. Consider the steady state graph of the process. Once the capacitor reaches V_{max}, it begins to discharge and after a time λ, it has a charge:

$$V_{min} = V_{max}\, e^{-\lambda/\tau} \quad (1)$$



Now capacitor can be broken in to a battery and an empty capacitor and charging begins, thus after a time λ, the potential difference:

$$V = V_\infty + (V_0 - V_\infty)\, e^{-\lambda/\tau} \Rightarrow V_{max} = V + (V_{min} - V)\, e^{-\lambda/\tau} \quad (2)$$

From (1) & (2), $V_{max} = \dfrac{V}{1 + e^{-\lambda/\tau}}$, $V_{min} = \dfrac{V}{1 + e^{\lambda/\tau}}$ Ans.

→ We can also find now max and min charges using $q = CV$

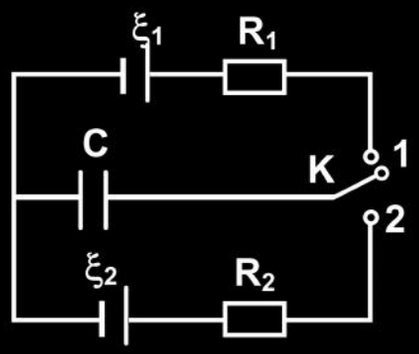
→ Current will periodically reverse its direction, Area under forward current must be equal to the area under reverse current (for current vs time graph).

→ We could use the same method even if periodically the voltage were following a square tooth wave between $V = V_1$ and $V = V_2$ and corresponding durations λ_1 and λ_2.

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EXAMPLE: KROTOV: 3.36:

The key K is connected alternately in turn to each of the contacts 1 and 2 over very short identical time intervals so that the change in the charge on the capacitor over each connection is small. What will be the final charge q_f on the capacitor?



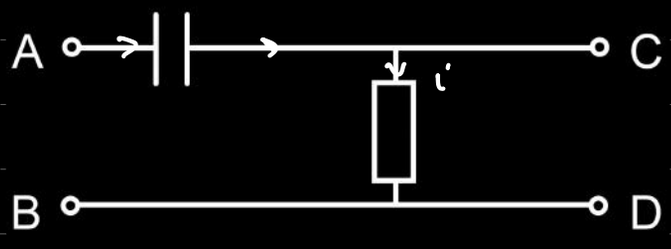
Solution: For charge to remain nearly constant, charge inflow during one connection must be equal to charge outflow during the other connection. Let steady state potential of capacitor be V, so we can break it in to a battery and uncharged capacitor. For visualization, let us assume $\xi_1 > \xi_2$, further let contact intervals be Δt each. Then inflow $(\xi_1 - V)\Delta t/R_1$, and outflow $(V - \xi_2)\Delta t/R_2$ thus for steady state: $(\xi_1 - V)\Delta t/R_1 = (V - \xi_2)\Delta t/R_2$

$$\Rightarrow V = \frac{\xi_1 R_2 + \xi_2 R_1}{R_1 + R_2} \text{ thus charge } q_f = C\left(\frac{\xi_1 R_2 + \xi_2 R_1}{R_1 + R_2}\right) \text{ Ans.}$$

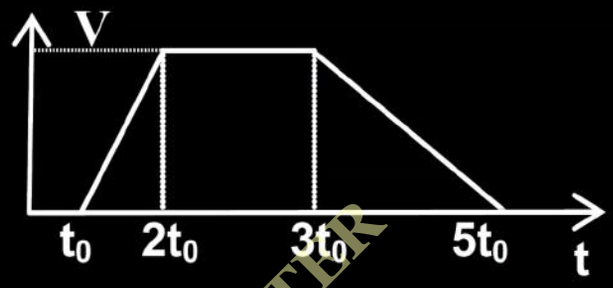
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EXAMPLE : KROTOV 3.42 :

A varying voltage is applied to the clamps AB such that the voltage across the capacitor plates varies as shown in Fig. Plot the time dependence of voltage across the clamps CD.



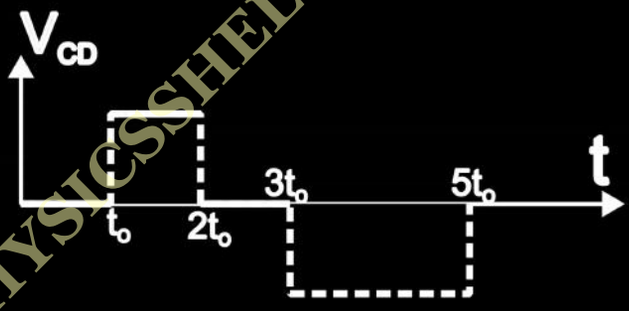
Draw only qualitative plots.



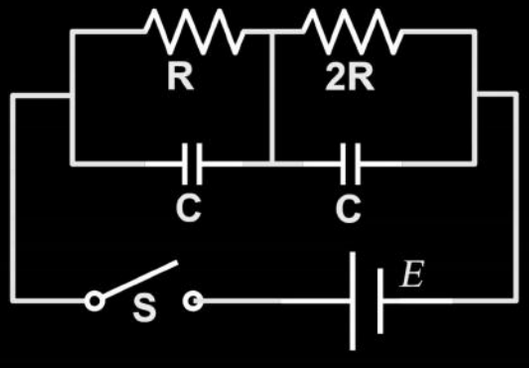
Solution: Capacitor and resistor are connected in series. Thus :

$$V_{CD} = iR = R\,dq/dt = R\frac{d}{dt}(CV)$$

$\Rightarrow V_{CD} = RC(dV/dt)$, thus we can infer V_{CD} just by looking at dV/dt of the capacitor voltage. This gives us the required graph of V_{CD} as shown in the adjacent figure.



RC CIRCUIT WITH ZERO RESISTANCE PATH AVAILABLE :



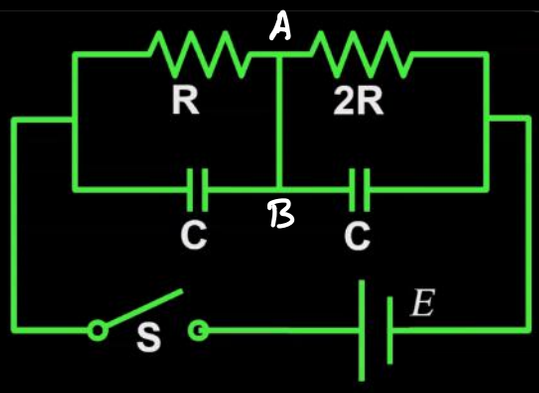
For finite potential differences any path with some finite resistance will have a finite charge flow rate, i.e. finite charge cannot suddenly flow through it, however for zero resistance paths sudden flow of charge is possible. For instance consider the adjacent circuit. Here to charge the capacitors, it is not necessary for charge to pass through any resistor. Thus immediately after closing the switch, the capacitors will get charged as if they were in series (If battery supplies a sudden charge, it must recieve the same charge back, but it can't get it through any resistor suddenly thus capacitors behave at initial moment as if they were in series. After this the charge flows and currents can be worked out by forming differential equations.

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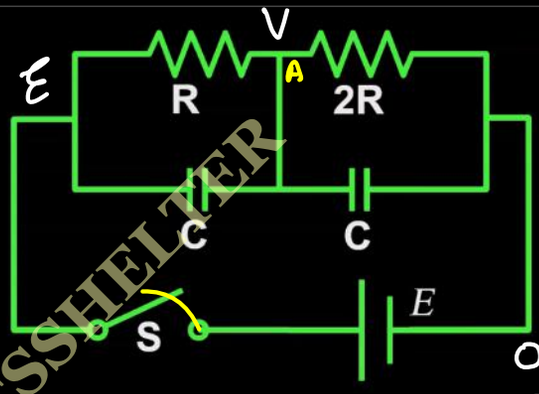
EXAMPLE: Consider the adjacent circuit. The switch is closed at $t=0$. Find:

(a) Charges on capacitor immediately after closing the switch.

(b) Potential of point A as a function of time, assuming zero potential at negative terminal of the battery.



Solution: As discussed in previous section, the capacitors will immediately get charged (as if in series). Thus initial potential drop across both capacitor is $\mathcal{E}/2$ each and their charges: $q_0 = C\mathcal{E}/2$ Ans.



Now we will find potential of point A using node voltage method. Applying junction law at node A:

$$\frac{V}{2R} + \left(\frac{V-\mathcal{E}}{R}\right) + (\dot{V}-0)C + \dot{V} \times C = 0 \qquad (\because \dot{\mathcal{E}}=0)$$

Thus $$2C\frac{dV}{dt} = -\frac{(3V-2\mathcal{E})}{2R} \Rightarrow \int_{\mathcal{E}/2}^{V} \frac{dV}{3V-2\mathcal{E}} = \int_0^t \frac{-1}{4RC}\,dt$$

Solving, we get: $$V = \frac{2\mathcal{E}}{3} - \frac{\mathcal{E}}{6}e^{-3t/4RC}$$ Ans

(Now if we want, we can find the charges and currents as a function of time using $V(t)$).

→ Here for finding capacitor charges at $t=0^+$, we treated resistors like open circuits.

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